Trust or Distrust in the Gaming Economy

Shared Practices and Missing Standards

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Gaming as an integral part of everyday social life

- The number of gamers worldwide is estimated to be over 3 billion (Newzoo 2024).
- Some surveys show 33% of adults aged 18 to 44 playing video games as leisure activity in a typical week (YouGov 2023).
- Gaming is part of the ambience of everyday life and an increasingly common form of social interaction (Richardson and Hjorth 2020).

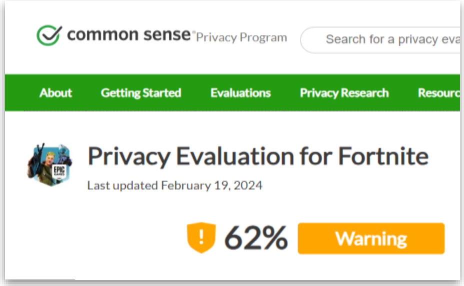
> "Before Instagram, Facebook, Twitter, and all the other platforms that we now umbrella under the term 'social media', sites for real-time social interaction via the internet already existed - computer games." (Taylor 2023: 1)

What are common data practices in gaming?

- Gaming platforms collect and process larger amounts of data (e.g., Kerr 2017, Nieborg 2015), prominently for commercial and capital purposes (Bernevega and Gekker 2019, Egliston and Padua 2023).

- However, research on user trust and privacy practices in gaming context is still limited (see Bourdoucen et al. 2023).

- Gaming research remains often not linked to the wider debates on the media landscape (Schwarzenegger et al. 2025).



Our objective: gaining a foundational understanding for trust and data literacy in the gaming context

- Landscape survey to understand common data practices across platforms
- Evaluation of three core practices identified in landscape survey
- Qualitative investigation on experience of data practices amongst gamers
- Quantitative study to evaluate levels of awareness, trust, and literacy amongst gamers

Three types of platforms surveyed in first project stage

📚 Publishing Platforms

… defined as the platforms on and through which people access games. Access here meaning purchasing games, managing libraries of game contents, downloading and installing game content to local devices, launching games through the platform clients.

Cases: Apple App Store, Epic Store, Google Play Store, Nintendo, PlayStation, Riot Games, Roblox, Steam, WeGame, Xbox.

🎮 Online Games

… defined as video games that are played online with and versus other players. Due to that, these games often involve the use of data intensive technologies, for example for social interaction, community moderation, or creating fair play conditions.

Cases: Apex Legends, Brawl Stars, Call of Duty, Clash of Clans, Counter-Strike, DotA2, FIFA, Fortnite, League of Legends, Minecraft, Mobile Legends, PUBG, Roblox, Rocket League, Valorant.

🥳 Community Sites

… defined as the platforms on which people connect as gamers with other gamers. This can include interactions with friends, but also public online communities, as well as through content and creators on other social media platforms.

Cases: Discord, Douyu, FACEIT, Facebook, Huya, KOOK, PerfectWorld Arena, Reddit, Twitch, YY, YouTube Gaming.

Privacy policies as core body of data

↓

Open coding of initial cases by each author

↓

Development of four key thematic groups

↓

Coding of all cases and secondary data
(e.g. blog posts, news articles, patents, etc.)

↓

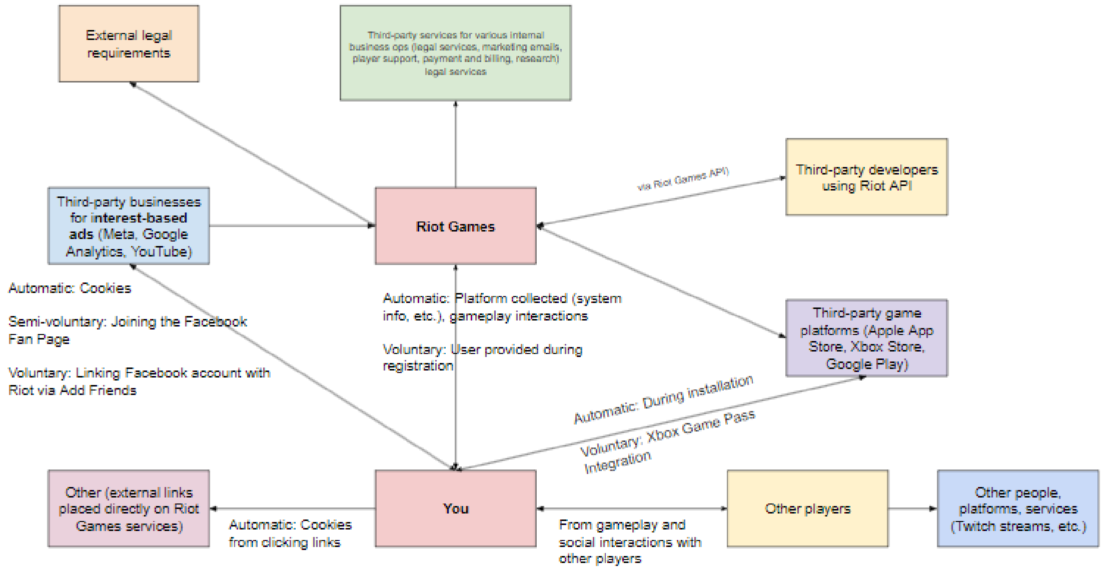
Development of themes + list of practices

Exploration of data subject positionalities

- Understanding trust as a relational construct (Sherman 2020), we investigate how gamers understand their position in the gaming landscape.

- The data-reliant nature of gaming (e.g., Kerr 2017) poses the question to how gamers negotiate their position of potential vulnerability in the trust relationship (Baier 1986).

Visualisation of data practices and relationships for the case of Riot Games (League of Legends)



Personalisation

- Content personalisation (e.g., based on usage, interests, localisation)
- **Personalisation of interactions** (e.g. game matchmaking systems)
- Friends and social recommendations (e.g. communities to join)
- Basic customisation, localization, and storage of preferences (e.g. via cookies technologies)

Moderation

- **Anti-cheat systems for fair play conditions**
- Various fraud detection (e.g. payment related)
- Anti-toxicity systems for ingame moderation
- Content detection, filtering, and moderation
- User reports and complaint processing
- Tools for community moderators

Advertising

- Hosting third-party tracking technologies on the platform
- Integrating into media delivery networks to host advertising content
- Sharing and receiving data about users from advertising networks

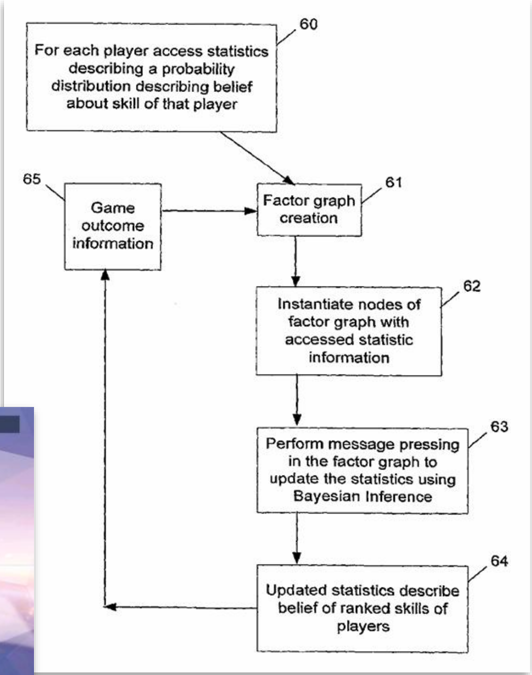
Matchmaking as personalisation

Matchmaking personalizes players' experience by shaping when, where, and with whom they can play against - through algorithms and data - primarily with the aim to create a "relevant" experience.

Decades long history:

- ELO (introduced in Chess)
- Skill-based (e.g. TrueSkill)
- Machine learning models
  - Device data
  - Behavioral scores
  - Skill assessments
  - Outcome estimations
  - Other types of data and models

Game Developer 2019 | Overwatch Matchmaking Screen



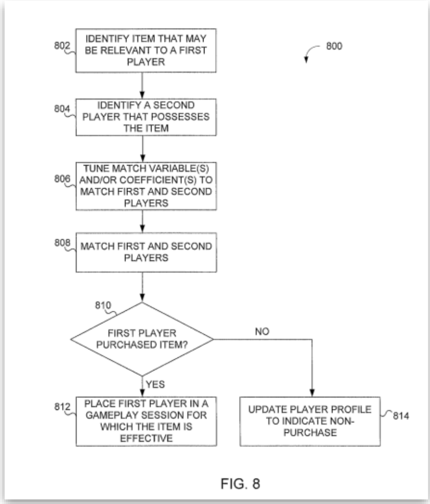
TrueSkill | WO2007094909A1

Ambivalence of trust in matchmaking

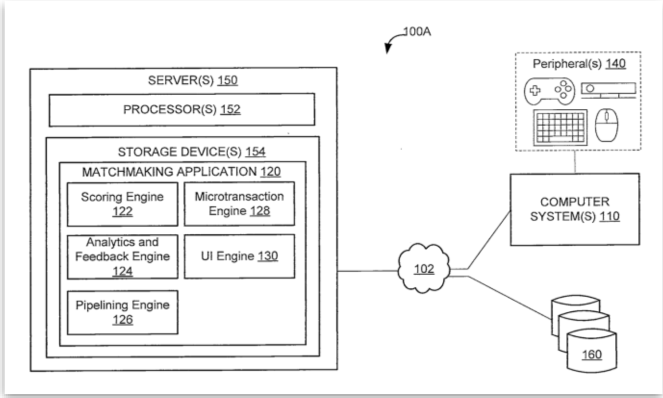
Expectancy violations

Unused patent by a major gaming publisher explored the process of matching players with those that have in-game items that the other looked at to encourage purchases.

Resulting controversy (PC Gamer 2017) and non-use of the algorithm showcase the ways in which boundaries and transgressions from trust in matchmaking are negotiated.



Unused Patent: US20160001181A1



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Personalisation folk theories on trust

Expectancy negotiation

Matchmaking system of a major online multiplayer game was suspected to enforce a 50/50 win ratio - i.e., when a player is on a loss streak the algorithm will try putting them on a team most likely to win the next match.

Players noticed patterns and a "folk theory" (DeVito et al. 2017) emerged, creating a moment of "digital irritation" (Ytre-Arne et al. 2021) in the community about the ways in which the algorithm impacts their player experiences.

MATCHMAKING UPDATES

- When a player is currently on a loss streak, the matchmaker will try to avoid putting the player on a team that is statistically calculated to have a lower chance of winning.

Developer comments: Loss streaks never feel good. Before each match, we make a prediction about which team will win the game, based on the information we have about the players on both teams. This is how modifiers such as Consolation, Reversal, Uphill Battle, and Expected give or take additional Rank Progress after each Competitive Play match. Since most matches will have a team that has a slightly higher chance of winning, placing a player on the team with the higher chance who is currently on a losing streak is aimed at helping them have a fair chance of breaking that streak. While this won't guarantee they'll win, it does provide a helping advantage.

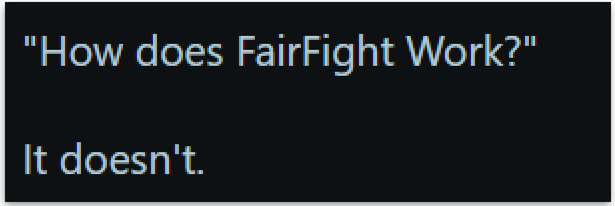
Overwatch 2 Retail Patch Notes – May 14, 2024

Anti-cheat as a form of moderation

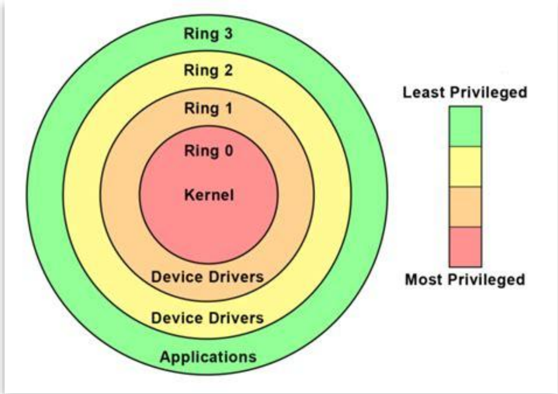
Anti-cheat are moderation practices and technologies that game developers deploy to enforce their terms of service and stop users from utilising (unapproved) third-party software in conjecture with the game software.

Long history and decades of development - parallel shift from human moderation (Game Masters) to data-driven and surveillance-based approaches (Kerr et al. 2014).

- Client side anti-cheat (monitors player devices on the kernel level for cheating software)
- Server side anti-cheat (anomaly detection and behavioural analysis via algorithms)



2016 Reddit Discussion - How FairFight works, what it does and what it doesn't do.



Ars Technica 2022

The complex ecologies of trust in moderation

Complex ecologies of trust (Steedman et al. 2020) arguing that trust is negotiated across layers of context. Vanguard Anti-Cheat Case:

Controversy around the technology emerged due to being kernel-based (accessing the most inner layer of a computing device).

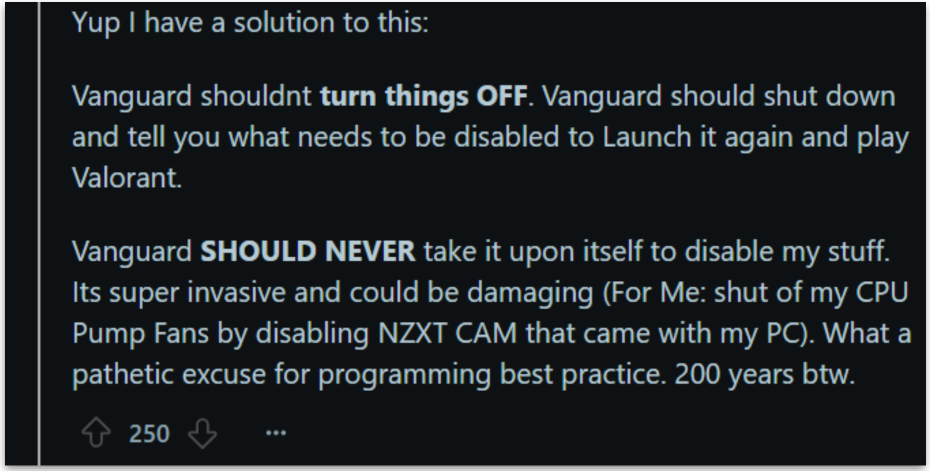
Trust negotiated not with technology itself, by the complex ecology of developer reputation, general risk from cyber attacks through the platform, and the personal gain from that technology.

> You expect a lot of trust from the players with this. But the thing is, trust in the technical department is hard to earn right now, with Riots recent data leak and the ever buggy client. Consider also the target you are making yourself: The biggest Game in the world, with unrestricted access to all its users data. You will be the first address for any and every hacker trying to make money. This can backfire terribly hard.
>
> And all the reason you give us is cheaters. I do not care about cheaters too much. I haven't seen a scripter in years, and when I watch streams of different elo brackets, cheaters are also very rare. I trusted Riot in the past to ban offenders through the report system and I thought they did a great job. If my games are being ruined, it is nearly always by tilted teammates who decide to lose, not by overperforming cheating enemies. I have next to zero gain from this, same as many others.

Reddit Discussion 2024 - Gameplay, Vanguard & More | Dev Update - League of Legends

The ambivalence of trust in anti-cheat technology

- Desire by communities to have anti cheat technologies - marker of trust in the developer to create a site for meaningful playful interaction.
- At same time, dominant technologies are invasive, carrying privacy and related effects (e.g., disruption of non-gaming experience of the computing device).
- Controversies and emotive reaction on these moderation practice showcase the intimacy of gaming as a data practice.



Yup I have a solution to this:

Vanguard shouldnt **turn things OFF**. Vanguard should shut down and tell you what needs to be disabled to Launch it again and play Valorant.

Vanguard **SHOULD NEVER** take it upon itself to disable my stuff. Its super invasive and could be damaging (For Me: shut of my CPU Pump Fans by disabling NZXT CAM that came with my PC). What a pathetic excuse for programming best practice. 200 years btw.

250 ...

Reddit Discussion 2020 - Vanguard does not tell you what is blocking, it just tells you that blocked something

The trust duality of personalisation and moderation

- Personalisation and moderation are desired data-driven practices that enhance gaming platform users' experiences.

  They provide a solution to the trust that users put in the platforms as sites for everyday playful interaction, free from abuse by cheaters or unbalanced competitive experience.

- At the same time, these data-driven practices create vulnerabilities in themselves that threaten trust and need to be negotiated.

  They provide further points for accessing gamers as commercial audiences, as well as interfere with people's everyday computing environment at the deepest level.

Trust is integral to understand how gaming platforms operate – and the controversial nature of data-driven technologies showcases both a fruitful field for research on technology trust yet also a need for higher attention in governance debates.

Thanks!

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